

TRIAD: Trusted RDF Interoperable Agent Data

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ABSTRACT

Patients can wait and deteriorate while information about them is missed, delayed, or scattered across records, and physicians spend much of their time piecing it together. We propose TRIAD, a data model that connects the information recorded by nurses and physicians so it can be shared and updated as it changes, and an agent can retrieve it directly rather than search for it. Code: github.com/MohamedKhalidmk/TRIAD (CC BY-NC-SA 4.0).

KEYWORDS

Knowledge Graph, FHIR, Agent Memory, Privacy

1 INTRODUCTION

When a patient calls a clinic, the information they give may be incomplete or inaccurate before a nurse summarizes it. Patients may be prioritized incorrectly, so urgent cases wait; a review of emergency triage reports nurse accuracy between 59.3% and 82% [10]. Physicians also lose time to wrong or outdated information in slow systems, spending 49% of their office day on electronic records and desk work [9].

Patient information comes from several people and systems over time: the patient's own report, the provincial pharmacy dispensing record (PharmaNet) [2], the nurse's assessment and recommendation (SBAR) [7], and the physician's decision, which may accept, reject, or build on that recommendation. Because this history is created by different people and organizations, records can disagree. The physician then cannot easily tell whether a record still reflects what the patient is actually taking [8], who recorded and verified it and when, and whether the nurse's recommendation was accepted.

2 MOTIVATION

Hospitals are slow to use agents in clinical work. Patient data sits in many separate systems, even where standards such as FHIR [5] are adopted, and an agent cannot hold all of it in its conversation window: in long contexts, language models miss relevant facts and act on outdated ones [6]. Electronic Health Records are designed for people to read, not to show an agent who asserted what, so its errors cannot be traced and staff stop trusting it.

Better-connected information would let patients be seen sooner and more safely and spare physicians time, when over half of Canadian physicians report high burnout and about half are considering reducing their clinical work [1].

FHIR gives systems a common structure for exchanging records and can note where each came from [4], but not a shared, machine-interpretable account of how records relate;

it points at conflicts rather than resolving them. Medication information is scattered across orders, dispenses, and patient reports, and clinicians reconcile it manually [3].

Patient information is also sensitive: identifying details linked to health data become Protected Health Information, governed by laws such as HIPAA, PIPEDA, and Ontario's PHIPA.

2.1 Motivating Example

At 9:30 a.m. on 8 July 2026, a patient comes to the UVic Health Services clinic with shortness of breath and tells the agent at intake that the problem started two days ago. The nurse records a possible asthma attack; one hour later, the physician revises the assessment to possible heart failure and tells the agent that the patient's father died of a heart attack, a fact not yet in the EHR. That evening, the patient asks the agent about the care plan. In the long conversation, the asthma assessment appears many times and the revision once, so the agent answers from the old one: an inhaler is the plan.

3 TEAM JUSTIFICATION

The project combines Mohamed Khaled's M.Sc. thesis research on memory for language model agents with Asish George's four years of experience developing healthcare systems. Mohamed Khaled wrote the abstract, introduction, and most of the motivation and set up the repository. Asish George wrote the motivation's opening and the example, named the project, designed its logo, and proposed FHIR. Both edited the proposal.

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A ORIGINAL WORK BEFORE AI-ASSISTED EDITING

Per the Level C guideline, this appendix contains both authors' original text, written without AI, before AI-assisted editing (A.1: Asish A. George, a spoken transcript; A.2 onward: Mohamed Khaled Abuela). AI (Claude) was also used to locate and explain sources.

A.1 Motivation: Original Transcript

Spoken transcript by Asish A. George.

I want to talk about the motivation for the project. There are different levels of motivation. First, in multi-agent systems, let's say a doctor has an agent to talk about patient data, and a nurse has an agent to talk about the patient data. These two agents may have fragmented memory about the patient data itself. We don't want different people to have fragmented memory about the same element. We want to have a cohesive memory within an agentic system so that patient care happens in a much easier way.

That aside, agentic systems are not being accepted or adopted into real healthcare systems as much, for different reasons. One is that the data is fragmented across different systems. The second is the concern of privacy, whether a data leak will happen. And third, there are strict regulations like HIPAA, and PHIPA in Canada, where we have to adhere to certain laws to support the healthcare environment.

We want to create a system or a data model that solves most of these problems. In an agentic system, we want a really fast lookup. So having a triple store, or RDF-based store, where we can query a patient's data would be beneficial for us.

The second is in terms of the data store. Like I said, there are multiple fragmented data stores, but the most common is FHIR data. It might not be widely adopted, but it is the global standard for interoperability for healthcare data. So we could model the agent in such a way that any FHIR data source can be attached to the agent, so that we can load FHIR data into the RDF store at any time, and the agent can operate within that data store.

The third is in terms of data privacy. Even though we are using a graph database to hold the relations, we also try to ensure that no data leaks, that one patient's data is not shown to another patient, and that only the practitioners who are responsible for the patient can see the patient.

This is the motivation behind the project and what we want to do with it.

A.2 Motivation: Who Benefits

everyone gains because its faster for the doctor and the patient saving living and not wasting time and easily doctors lives when doctors are in most need and getting there jobs

A.3 Motivation: FHIR Structure vs. Meaning

yeah fhir is for common structures ours is for common machine readable structures

A.4 Abstract

patient wait and goes critical and miss information keeps delaying while physicians is working 24/7 and we are just trying to make standards that arent helping but just pointing triad automates the entire processes from summary to giving the doctor what he needs using a smart data model that doesnt require the agent to ask questions and fetch but just to pull without reason sharing across nurse and doctor info can be shared and exchanged and edited right away without any problems all connected

A.5 Problem Definition Notes

Answers to the six problem-definition questions, as originally written:

1. patient on phone then pharmacy then xrays then ISMP then SBAR
2. patient history and medications different other clinics or out of data, also name different in entities like m.khaled or mohamed khaled or different descriptions of symptoms
3. what patient who wrote it who verified
4. Next patient, condition, how urgent and what they are currently taking and nurse recommendation
5. same
6. why it was recorded who recorded it when and patient full history and condition

Follow-up answers: medications the patient currently taking and it gives the doctor an idea and the doctor can use it or reject it or expand it; history and results; human symbols and writing and also what patient and and time and author revision verification etc.

Problem paragraph: Patient history can differ as its created by different people and organisation and its not easy to know without wasting the doctors time also a nurse can miss important info over the phone and the doctor can waste some time on something he didnt need to do while also other more important patient suffer

A.6 Introduction: Opening

so the patient when he calls a hospital the info he give might not be that accurate or summaries so the nurse summaries it [...] that has some serious problems which are first the ordering of patients are incorrect, causing important patients to wait critical time and see that the doctor time is wasted with completely wrong info, also doctors deal with the slowest outdated systems that doesnt just give the info right away we need a source for that

A.7 Introduction: Disagreement Between Records

Because this history is created by different people and organizations, records can disagree, and the same patient may appear under different names. When records disagree, the physician cannot easily tell whether a record still reflects what the patient is actually taking, who recorded it and when, who verified it, and whether the nurse's recommendation was accepted, rejected, or built on.

A.8 Introduction: Gap

fhir just signaties what comes from a different source not resolve the conflict its pointing at the problem not offering solvtions and also the sattering thing i said and also most of clinical work is still maual and u can find many evidence